

# 금융투자론 2

2025-1

Returns and Risk

# Returns and Risk

2

- **OBJECTIVE:** to understand the components of required return from an investment. We learn how to measure average/expected returns and return standard deviation. We also investigate the risk premiums provided by several key assets from the historical data.

# Motivation

3

- Finance is about “Valuation”.
- Two Basic principles in finance (valuation):
  - 1. Time Value of Money: discount
  - 2. Risk and return
- The **required return** reflects these two principles, and plays a crucial role in valuation.
  - ▣ The future cash flows will be **discounted** by using this rate.
  - ▣ This note’s main theme is how the required return should be **formulated**.

# Required Rate of Return

4

- The **minimum** rate of return that an investor wants from an investment to **compensate her/him for deferring consumption**.

- As an opportunity cost

- As a financing cost

- Components of Required Rate of Return

$$\begin{aligned}\text{Req. Return} &= (1+RR_f)(1+E(\pi))(1+RP) - 1 \\ &\approx RR_f + E(\pi) + RP \\ &= R_f + RP\end{aligned}$$

where

$RR_f$  = Real Risk-free Rate

$E(\pi)$  = Expected Rate of Inflation

$RP$  = Risk Premium

$R_f$  = Nominal Risk-free Rate

(+)

$R_f$  ,  $RP \rightarrow RP$  ?

# Measuring a Rate of Return

5

- Suppose you bought 100 shares of Home Depot for \$40 per share a year ago. You sold all the shares today for \$60. You have received \$0.50 per share dividends over the year.

- Dollar Returns

$$\begin{aligned}\text{Total return} &= \text{Dividend Income} + \text{Capital Gains} \\ &= 0.5 \times 100 + (60 - 40) \times 100 = \$2,050\end{aligned}$$

- Percentage Returns

$$\begin{aligned}\% \text{ return} &= (\text{Dividend Income} + \text{Capital Gains}) / \text{Investment} \\ &= \text{Dividend yield} + \text{Capital gains yield} \\ &= 1.25\% + 50\% = 51.25\%\end{aligned}$$

50/4000

2000/4000

# Holding Period Return

6

- Given the yearly returns you can calculate the holding period return by

$$\text{Holding period return} = (1 + R_1) (1 + R_2) \dots (1 + R_t) - 1$$

- Why not  $R_1 + R_2 \dots + R_t$ ? Over-valued

|      |       |     |     |    |     |
|------|-------|-----|-----|----|-----|
| (Ex) | Year  | 0   | 1   | 2  | 3   |
|      | Price | 100 | 150 | 75 | 100 |

+50%

-50%

+33%



# Return Statistics

7

- What are statistics?
  
- Measures of Average Return: **mean**
  - ▣ Using historical data
  - ▣ Using probabilistic assumptions
  
- Measures of Return Volatility: **variance**
  - ▣ Using historical data
  - ▣ Using probabilistic assumptions
  
- Measures of Covariability: **covariance**
  - ▣ Using historical data
  - ▣ Using probabilistic assumptions

# Return From Historical Data

8

- Given the series of historical (e.g. yearly) returns, return statistics intend to summarize these series of numbers

## 1. Average return

- **Geometric Mean:**  $r_g = \{[(1+r_1) (1+r_2) \dots (1+r_n)]\}^{1/n} - 1$

Average holding period return assuming **compounding**

=> More **relevant**

- **Arithmetic Mean:**  $r_a = (r_1 + r_2 + \dots + r_n) / n$

Simple Average assuming no compounding

=> Easy to use, but has **upward bias**





# Return From Historical Data cont.

9

## □ Example

Calculate the arithmetic mean return of the following company. Assume that the company does not pay any dividend.

| Year                               | 1991  | 1992  | 1993  | 1994  | 1995  |
|------------------------------------|-------|-------|-------|-------|-------|
| Price at the beginning of the year | \$100 | \$110 | \$120 | \$130 | \$120 |
| Return                             |       | 0.10  | 0.09  | 0.08  | -0.08 |
| Arithmetic mean return             | ?     |       |       |       |       |

## □ What is the **geometric** mean return?

$$r = \{(1.1)(1.09)(1.08)(0.92)\}^{1/4} - 1 = 4.47\%$$

Note that these are rounded-up

# Return From Historical Data *cont.*

10

## □ Bias in Arithmetic Mean Return

There is an upward bias in arithmetic mean return.

|       | 2017        | 2018 | 2019 |                      |
|-------|-------------|------|------|----------------------|
| (Ex.) | stock A 100 | 150  | 100  | <input type="text"/> |
|       | stock B 100 | 50   | 100  |                      |

What are the arithmetic mean returns?

As the price series gets more volatile this bias becomes larger.

Try with (100, 200, 100)

# Return Variance From Historical Data

11

## 2. Return Variance: measure for volatility (risk) ☐

$$\sigma^2 = \frac{1}{N-1} \sum_{t=1}^N (r_t - \bar{r})^2, \quad \text{where } N \text{ is the number of observations, } r_t \text{ is the return of the stock at time } t, \text{ and } \bar{r} \text{ is the average return during the sample period.}$$

### □ Standard deviation ( $\sigma$ ) = (Variance)<sup>1/2</sup>

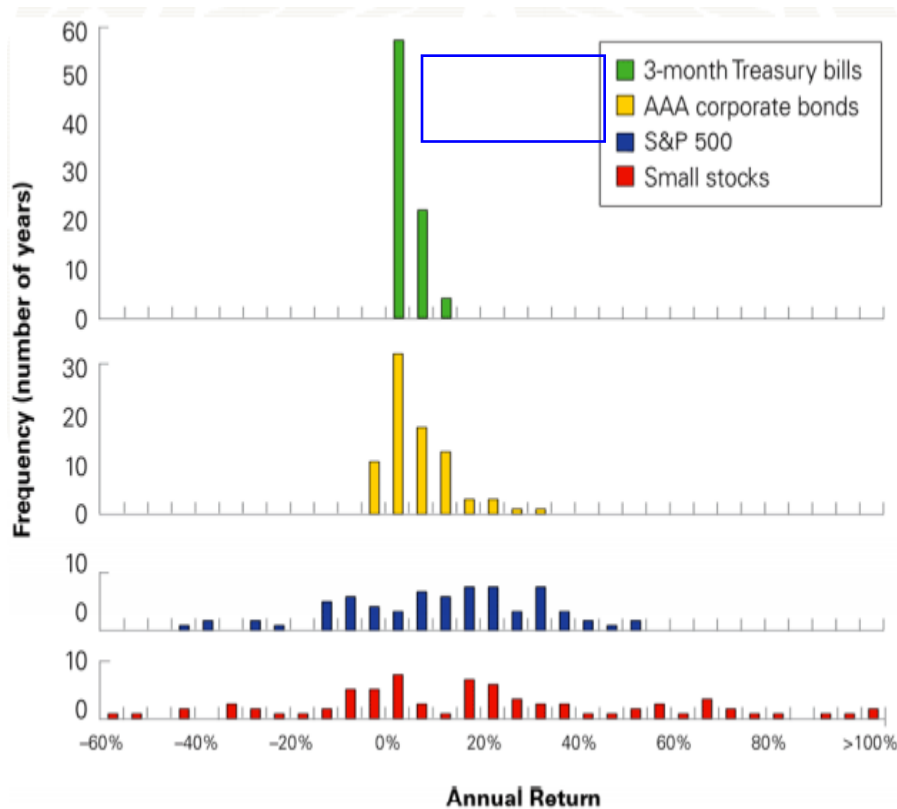
(Ex) Calculate the var. and STD from the previous example

$$\text{Var} = 0.0073 = (1/3) \times \{(0.1-0.0447)^2 + (0.09-0.0447)^2 + (0.08-0.0447)^2 + (-0.08-0.0447)^2\}$$

$$\text{STD} = 0.0073^{1/2} : 8.54\%$$

# U.S. market returns, 1926-2008

12



| Investment      | Average Annual Return |
|-----------------|-----------------------|
| Small stocks    | 20.9%                 |
| S&P 500         | 11.6%                 |
| Corporate bonds | 6.6%                  |
| Treasury bills  | 3.9%                  |

| Investment      | Return Volatility (Standard Deviation) |
|-----------------|----------------------------------------|
| Small stocks    | 41.5%                                  |
| S&P 500         | 20.6%                                  |
| Corporate bonds | 7.0%                                   |
| Treasury bills  | 3.1%                                   |

# Covariance of Returns From Historical Data

13

## 3. Covariance of Returns

$$Cov_{1,2} = \frac{1}{N} \sum_{t=1}^N (r_{t,1} - \bar{r}_1) \times (r_{t,2} - \bar{r}_2)$$

(Ex)

| Year         | Stock 1 | Stock 2 |
|--------------|---------|---------|
| 1995         | 0.10    | 0.20    |
| 1996         | -0.15   | -0.20   |
| 1997         | 0.20    | -0.10   |
| 1998         | 0.25    | 0.30    |
| 1999         | -0.30   | -0.20   |
| 2000         | 0.20    | 0.60    |
| Average Ret. | 0.05    | 0.10    |

Verify  
that

Cov =  
0.0425

$$r_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

$$E[(X - \mu_X)(Y - \mu_Y)] = \frac{\sum_{i=1}^m (X_i - \mu_X)(Y_i - \mu_Y)}{m},$$

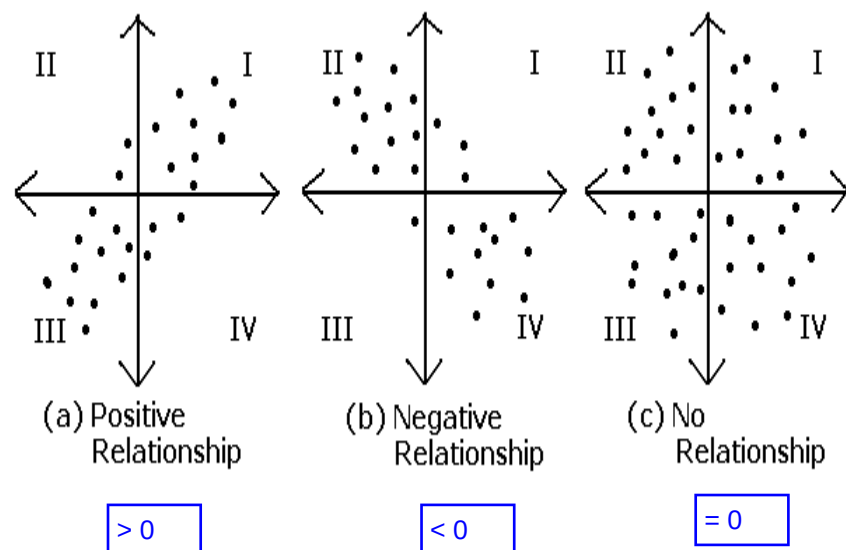
$\mu_X$  : 모집단  $X$ 의 평균

$\mu_Y$  : 모집단  $Y$ 의 평균

$\sigma_X$  : 모집단  $X$ 의 표준편차

$\sigma_Y$  : 모집단  $Y$ 의 표준편차

$m$  : 모집단의 개체수



# Expected Return

## From Probabilistic Assumptions

15

- Given a probability distribution over possible states, and return on the stock as a function of the state,

$$E(R) = \sum_{i=1}^s p_i \times r_i, \quad \text{where}$$

*S is the number of states,  $p_i$ , prob. of state  $i$ , and  $r_i$ , return in state  $i$ .*

- (Ex)

| State of Economy  | Probability of state <sub><i>i</i></sub> ( $p_i$ ) | Return in state <sub><i>i</i></sub> ( $r_i$ ) |
|-------------------|----------------------------------------------------|-----------------------------------------------|
| +1% change in GDP | .25                                                | -5%                                           |
| +2% change in GDP | .50                                                | 15%                                           |
| +3% change in GDP | .25                                                | 35%                                           |

$$E(R) = \{(.25) \times (-5) + (.50) \times (15) + (.25) \times (35)\} = -1.25 + 7.50 + 8.75 = 15\%$$

# Variance of Returns

## From Probabilistic Assumptions

16

- Given the expected return,

$$\text{Var}(R) \equiv \sigma^2 = \sum_{i=1}^S p_i \times (r_i - \bar{r})^2, \quad \text{where } \bar{r} \text{ is the expected return.}$$

- From the previous example,

| i   | $(r_i - \bar{r})^2$ | $p_i \times (r_i - \bar{r})^2$ |                                |
|-----|---------------------|--------------------------------|--------------------------------|
| i=1 | .04                 | .01                            | $0.25 \times (-0.05 - 0.15)^2$ |
| i=2 | 0                   | 0                              |                                |
| i=3 | .04                 | <u>+.01</u>                    |                                |
|     |                     | $\text{Var}(R) = .02$          |                                |

What is the standard deviation?



# Covariance of Returns

## From Probabilistic Assumptions

17

- Given probabilities of certain states with returns of the stocks as functions of the states, first evaluate the expected returns of the two stocks, and

$$Cov(R_1, R_2) \equiv \sigma_{12} = \sum_{i=1}^S p_i (r_{1i} - \bar{r}_1)(r_{2i} - \bar{r}_2),$$

where  $\bar{r}_i$  is the expected return on stock  $i = 1, 2$

| State of the economy | Probability of state | Return on asset 1 | Return on asset 2 |
|----------------------|----------------------|-------------------|-------------------|
| Boom                 | 0.40                 | 30%               | -5%               |
| Bust                 | 0.60                 | -10%              | 25%               |

# Covariance of Returns

## From Probabilistic Assumptions *cont.*

18

### Step 1. Expected returns

$$E(R_1) = 0.4 \times (0.30) + 0.6 \times (-0.10) = 0.06 (= 6\%)$$

$$E(R_2) = 0.4 \times (-0.05) + 0.6 \times (0.25) = 0.13 (= 13\%)$$

### Step 2. Covariance of returns

$$\begin{aligned} \text{Cov}(R_1, R_2) &= 0.4(0.3-0.06)(-0.05-0.13) \\ &\quad + 0.6(-0.1-0.06)(0.25-0.13) \\ &= -0.0288 \end{aligned}$$

